YIKANG OUYANG

The Hong Kong University of Science and Technology (Guangzhou)
PhD ◇ Microelectronics Thrust ◇ Personal Website
PhD Advisor: Prof. Yuzhe Ma
youyang929@connect.hkust-gz.edu.cn

RESEARCH INTERESTS

- Electronic Design Automation (EDA), ML-aided VLSI design.

EDUCATION

The Hong Kong University of Science and Technology (Guangzhou), China Aug. 2022 – Oct. 2026
Ph.D. Microelectronics

Sun Yat-Sen University, Guangzhou, China Sep. 2018 – Jul. 2022
B.Eng. Microelectronics

AWARDS AND HONORS

[A4] MLCAD Student Travel Grant, 2026.

[A3] **Best Paper Award Nomination**, IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC), 2025.

[A2] **Best Paper Award**, ACM/IEEE International Workshop on Machine Learning for CAD (MLCAD), 2023.

[A1] Full Postgraduate Scholarship, HKUST(GZ), 2022–2026.

PROJECTS

Logic Synthesis Modeling & Optimization

- Proposed an efficient gradient-based optimization framework to optimize logic synthesis parameters.
- Proposed a ML model to accurately predict the delay and area given RTL design and synthesis parameters.
- Publications: MLCAD'23 [Best Paper Award], DAC'25

Power Modeling & Optimization

- Proposed an efficient gate-level glitch power estimation method by sampling simulation traces.
- Proposed a sensitivity analysis method to optimize glitch power by gate sizing.
- Publications: ASP-DAC'25 [Best Paper Nomination], TACO'26, TCAD (SMART-GPO 2.0, just accepted)

Standard Cell Characterization

- Proposed a fast and accurate standard cell library characterization method with physics-informed ML model.
- Publications: On-going project

LLM for Chip Design Automation

- Proposed an LLM multi-agent framework for automatic chip PPA optimization through engineering-change-order (ECO).
- Publications: MLCAD'26

Mask Optimization in VLSI Manufacturing

- Use RL to optimize lithographic mask.
- Publications: TCAD'24

Datapath Design Space Exploration

- Design multipliers with superior compressor tree structures.
- Publications: DAC'23, TODAES'26

PUBLICATIONS

Journal Papers

[J4] **Yikang Ouyang**, Yuchao Wu, Dongsheng Zuo, Subhendu Roy, Tinghuan Chen, Zhiyao Xie, Yuzhe Ma, "SMART-GPO 2.0: Gate-Level Enhanced Sensitivity Measurement with Fast and Accurate Estimation for Glitch Power Optimization", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (**TCAD**), just accepted.

[J3] Weihao Wang, **Yikang Ouyang**, Hongyuan Liu, Yuzhe Ma, "GPU-Accelerated Gate-Level Time-Based Power Analysis via Event-Density-Aware Partitioning and Kernel Fusion", ACM Transactions on Architecture and Code Optimization (**TACO**), 2026.

[J2] Dongsheng Zuo, Jiadong Zhu, **Yikang Ouyang**, Yuzhe Ma, "RL-MUL 2.0: Multiplier Design Optimization with Parallel Deep Reinforcement Learning and Space Reduction", ACM Transactions on Design Automation of Electronic Systems (**TODAES**), vol. 31, no. 4, pp. 1–20, 2026.

[J1] Xiaoxiao Liang, **Yikang Ouyang**, Haoyu Yang, Bei Yu, Yuzhe Ma, "RL-OPC: Mask Optimization with Deep Reinforcement Learning", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (**TCAD**), vol. 43, no. 01, pp. 340–351, 2024.

Conference Papers

$*$ denotes equal contribution.

[C5] **Yikang Ouyang**, Yang Luo, Dongsheng Zuo, Yuzhe Ma, "Retrieve, Schedule, Reflect: LLM Agents for Chip QoR Optimization", ACM/IEEE Workshop on Machine Learning for CAD (**MLCAD**), Jeju, 2026.

[C4] **Yikang Ouyang***, Xiaofei Yu*, Jiadong Zhu, Tinghuan Chen, Yuzhe Ma, "Efficient Continuous Logic Optimization with Diffusion Model", 62nd ACM/IEEE Design Automation Conference (**DAC**), pp. 1–7, San Francisco, Jun. 2025.

[C3] **Yikang Ouyang**, Yuchao Wu, Dongsheng Zuo, Subhendu Roy, Tinghuan Chen, Zhiyao Xie, Yuzhe Ma, "SMART-GPO: Gate-Level Sensitivity Measurement with Accurate Estimation for Glitch Power Optimization", IEEE/ACM Asia and South Pacific Design Automation Conference (ASP-DAC), Tokyo, Jan. 2025.**(Best Paper Award Nomination)**

[C2] **Yikang Ouyang**, Sicheng Li, Dongsheng Zuo, Hanwei Fan, Yuzhe Ma, "ASAP: Accurate Synthesis Analysis and Prediction with Multi-task Learning", ACM/IEEE Workshop on Machine Learning for CAD (**MLCAD**), Utah, Sep. 2023. **(Best Paper Award)**

[C1] Dongsheng Zuo, **Yikang Ouyang**, Yuzhe Ma, "RL-MUL: Multiplier Design Optimization with Deep Reinforcement Learning", ACM/IEEE Design Automation Conference (**DAC**), San Francisco, Jul. 09-13, 2023.

SKILLS

- Programming Language: Python, C/C++, MATLAB
- Hardware Design Language: Verilog, CHISEL
- AI Toolkits: PyTorch, LangChain, LangGraph
- VLSI Design Software: VCS, Design Compiler, Innovus, ICC2, PrimeTime, HSPICE